Transition-Aware Boundary-Guided Dual-Frequency Network for Hyperspectral Unmixing

Yaxiong Chen, *Member, IEEE*, Siqi Liang, Quanchuan Cheng, Bo Zhang, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*—Hyperspectral unmixing (HU) aims to decompose mixed pixels into endmembers and their corresponding abundances. However, material transition zones often suffer from severe spectral mixing and error diffusion, making abundance estimation particularly brittle near boundaries. To substantiate this claim, we first formulate and validate a boundary–transition prior through a testable hypothesis. High-frequency (HF) responses extracted from observed hyperspectral images (HSIs) consistently align with material boundaries. Quantitative alignment evidence on synthetic data (with abundance ground truth) and real scenes (using structural boundary proxies) supports this prior. Building on this validated prior, we propose a transition-aware boundary-guided dual-frequency network for HU. A stationary wavelet transform (SWT) decomposes the HSI into low-frequency (LF) and HF components that are modeled by dual branches, while a learnable boundary-guidance map predicted from the HF branch adaptively fuses dual-frequency representations to refine transition regions without imposing explicit spatial smoothness regularization. A lightweight global guidance pathway further enriches spectral–spatial context. Experiments on synthetic and real datasets demonstrate that the proposed TABG-Net significantly outperforms state-of-the-art methods, especially in preserving sharp boundaries and producing accurate abundances in transition-dominated scenarios. The code is publicly available at https://github.com/hgxcxdg/TABG-Net**

***Index Terms*—Autoencoder (AE), boundary-guided learning, deep learning (DL), hyperspectral unmixing (HU), stationary wavelet, transition aware.**

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Yaxiong Chen is with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, and also with Hubei Key Laboratory of Transportation Internet of Things, School of Artificial Intelligence, Wuhan University of Technology, Wuhan 430072, China.

Siqi Liang is with the School of Computer Science, Shanghai Jiao Tong University, Shanghai 200240, China.

Quanchuan Cheng and Bo Zhang are with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, also with the School of Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China.

Shengwu Xiong is with Hubei Institute for Heterogeneous Computing Applications, Wuhan College, Wuhan 430212, China (e-mail: xiongsw@whut.edu.cn).

Xiaoqiang Lu is with the College of Physics and Information Engineering, Fuzhou University, Fuzhou 350108, China.

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I. Introduction

HYPERSPECTRAL image (HSI), which integrates imaging spectroscopy and remote sensing technologies, offers high spectral resolution and enables the simultaneous acquisition of spatial and spectral information [1]. Owing to these advantages, HSI has been widely applied in various fields, including environmental monitoring [2], medical diagnosis [3], mineral analysis [4], military reconnaissance [5], image classification [6], [7], [8], and target detection [9], [10]. By capturing hundreds of contiguous narrow spectral bands ranging from the visible to short-wave infrared regions [11], HSIs provide detailed spectral signatures and complex spatial structures of surface materials [12]. Therefore, spatial and spectral information often needs to be considered together [13]. However, owing to the inherent tradeoff between the spectral and spatial resolutions of imaging sensors, combined with the high complexity of surface material distributions, HSI inevitably contain mixed pixels, which comprise multiple constituent materials. This spectral mixing effect significantly impairs the effectiveness of HSI in fine-grained material identification, posing challenges for accurate analysis and interpretation [14].

To address this problem, hyperspectral unmixing (HU) has emerged as a critical technique. Its objective is to decompose each mixed pixel at the subpixel level into a set of spectrally pure components, known as endmembers, and to estimate their corresponding fractional abundances [15]. The unmixing process relies heavily on the modeling of the spectral mixing mechanism. Existing methods can be broadly categorized into linear mixing model (LMM) and nonlinear mixing model (NLMM) [16]. The LMM assumes that the observed spectrum of a pixel is a linear combination of multiple endmember spectra, weighted by their abundance fractions [17]. Owing to their simplicity and strong generalizability, LMMs have been widely adopted in practice. In contrast, the NLMM aims to capture the nonlinear interactions of photons with surface materials more accurately [18], such as multiple scattering and complex reflectance effects. Although more realistic theoretically, NLMM typically requires a larger number of parameters

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and prior knowledge [19], thereby increasing the complexity and difficulty of model construction.

Traditional HU procedures are typically divided into two stages: endmember extraction and abundance estimation. However, errors introduced during endmember extraction are likely to propagate and adversely affect the accuracy of abundance estimation. To mitigate this issue, recent studies have proposed joint estimation approaches that simultaneously estimate endmembers and abundances [20]. One representative method is nonnegative matrix factorization (NMF) [21], which incorporates additional constraints, such as sparsity or spatial consistency, into the decomposition process to enhance the unmixing performance [22]. Nevertheless, the design of such constraints not only increases computational complexity but also often relies on strong prior knowledge, posing significant challenges in practical applications.

In recent years, deep learning (DL)-based methods have gained increasing attention in the field of HU due to their powerful representation capabilities and ability to model complex nonlinear relationships [23], [24]. Unlike traditional model-driven approaches, DL-based methods adopt a data-driven paradigm to learn hierarchical spectral–spatial features directly from hyperspectral data [25]. Guo et al. [26] first applied autoencoders (AEs) to HU, marking the early adoption of DL in this domain. Subsequently, Su et al. [27] mathematically demonstrated the suitability of AE structures for HU in their DAEN paper, providing a solid theoretical foundation for the widespread adoption of AE-based unmixing methods. Since then, AE-based unmixing has gradually become the mainstream approach in the unmixing literature. By leveraging convolutional neural networks (CNNs) [28], AEs [29], and attention mechanisms [30], these methods have demonstrated superior performance in extracting endmembers and estimating abundances, particularly in scenarios with complex mixing patterns and low signal-to-noise ratios (SNRs). Moreover, the integration of prior knowledge, such as sparsity and spectral variability, into network architectures or loss functions has further improved the robustness and accuracy of unmixing results [31]. As a result, DL-based unmixing approaches are emerging as a promising alternative to conventional techniques. For instance, Vijayashekhar et al. [32] proposed a two-stage fully connected self-supervised DL network to address practical issues in blind HU. The network jointly estimates endmembers and abundances in the first stage and learns the physics of hyperspectral image acquisition in the second stage, automatically learning the unmixing function from the sensed hyperspectral data, even at lower SNRs. Palsson et al. [33] proposed a DL-based method for blind HU using a neural network AE. The method shows that the linear mixture model imposes implicit architectural constraints on the network and performs unmixing effectively. Su et al. [34] proposed MAAENet, a multiattention AE network for HU based on the extended LMM. The method addresses spectral variability in real scenes by incorporating a spatial–spectral attention (SSA) module and a sparse constraint based on spatial homogeneity, improving the extraction of global spatial features and handling band redundancy. Jin and Yang [35] proposed a graph attention convolutional AE for HU,

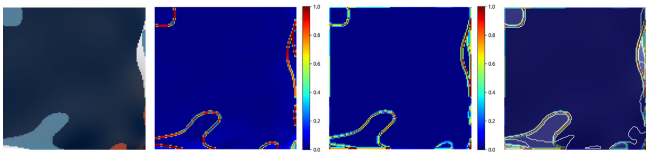


Fig. 1. Spatial motivation of the wavelet-based boundary prior (synthetic data with abundance ground truth). (a) Pseudo-RGB visualization of the hyperspectral image. (b) GT abundance-boundary strength map $\|\nabla \mathbf{A}\|$, normalized to $[0, 1]$; the boundary contour is obtained by thresholding the top 10% responses. (c) Spatial HF energy map extracted by SS-1D-SWT, normalized to $[0, 1]$. (d) Overlay of (c) with the contour from (b), showing that the SS-1D-SWT HF responses tend to highlight material transition boundaries.

addressing the limitation of existing AE-based methods in fully utilizing spatial information. By incorporating graph attention convolution, the method enhances the extraction of both long-range and short-range spatial features, improving abundance estimation with global and local spatial consistency. The decoder is further improved using the postpolynomial NLMM, which enhances the physical interpretability for nonlinear blind unmixing. Jin et al. [36] proposed a multilinear HU method based on an AE network and recurrent neural network (RNN), addressing the limitations of existing nonlinear models that rely on bilinear mixing. The method employs an RNN in the decoder to simulate multiple light scattering events and adaptively updates the RNN weights during training, providing a physical interpretation of the nonlinear interactions between materials.

Moreover, the integration of diverse and informative features has been recognized as a key factor for improving the performance of DL-based HU methods. Beyond traditional spectral information, many studies have incorporated external data sources to enhance structural awareness. For example, integrating LiDAR data provides precise 3-D spatial cues, such as height and shape, to help distinguish spectrally similar materials [37]. Other research utilizes multimodal fusion or multiscale perception blocks to capture robust spectral–spatial features [24], [38], [39], [40], [41]. However, while these auxiliary sources significantly enhance model capability, their acquisition is often constrained by sensor availability and cost in practical applications. This limitation motivates the need for methods that can extract equivalent, boundary-aware structural priors directly from the internal characteristics of the HSI itself.

Unfortunately, most existing deep HU methods primarily emphasize holistic spectral modeling and often underexploit local structural cues, particularly in transition zones between adjacent materials where mixed pixels and spectral ambiguity are more prevalent [42], [43], [44]. These transition zones are particularly challenging for accurate unmixing because they contain a higher proportion of mixed pixels and exhibit sharp abundance transitions near material boundaries. Although spatial smoothness or consistency regularization is frequently introduced to stabilize abundance estimation [45], [46], it can inadvertently oversmooth these discontinuous regions, blur material boundaries, suppress minority components, and propagate errors across neighboring pixels, thereby degrading edge fidelity and making HU predictions brittle near boundaries.

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TABLE I

BOUNDARY/TRANSITION PRIOR VALIDATION ON A SYNTHETIC HSI (GT AVAILABLE). WE EVALUATE THE ALIGNMENT BETWEEN THE HF MAPS PRODUCED BY DIFFERENT DECOMPOSITIONS AND THE ABUNDANCE-BOUNDARY GROUND TRUTH DEFINED BY THE TOP 10% PIXELS OF $\|\nabla A\|$. THE BEST RESULTS ARE IN BOLD. RUNTIME IS REPORTED AS MEDIAN TIME (MS)

Method	Spearman ↑	ROC-AUC ↑	PR-AUC ↑	P@top10% ↑	R@top10% ↑	Runtime (median, ms) ↓
SS-1D-SWT	**0.8077**	0.9509	**0.7227**	**0.6752**	**0.6748**	**119.15**
2D-SWT	0.7903	**0.9513**	0.7106	0.6264	0.6260	135.44
1D-DWT	0.7078	0.9340	0.6787	0.5971	0.5967	131.39
FFT-HP	0.3028	0.8584	0.5372	0.5733	0.5729	141.36

TABLE II

BOUNDARY ALIGNMENT ON REAL HSIS USING A STRUCTURAL BOUNDARY PROXY (SOBEL GRADIENT ON THE FIRST PCA COMPONENT, WITH THE TOP 10% PIXELS AS THE BOUNDARY MASK). HIGHER IS BETTER FOR ALL METRICS. RUNTIME IS REPORTED AS MEDIAN TIME (MS)

Dataset	Method	Spearman ↑	ROC-AUC ↑	PR-AUC ↑	P@top10% ↑	R@top10% ↑	Runtime (median, ms) ↓
Samson	SS-1D-SWT	0.7551	0.8342	0.2896	0.3548	0.3544	28.50
	2D-SWT	**0.7682**	**0.8563**	**0.3223**	**0.3858**	**0.3854**	35.95
	1D-DWT	0.6877	0.8237	0.2623	0.3215	0.3212	**25.70**
	FFT-HP	0.3033	0.6579	0.1566	0.2029	0.2027	63.26
Houston	SS-1D-SWT	0.5672	0.8789	0.5473	0.5218	0.5213	28.31
	2D-SWT	**0.5817**	**0.8864**	**0.5491**	**0.5327**	**0.5322**	35.52
	1D-DWT	0.5594	0.8666	0.5261	0.5136	0.5131	**24.69**
	FFT-HP	0.4035	0.7962	0.4468	0.4528	0.4524	36.99

These observations suggest that accurate unmixing in transition zones requires an explicit, boundary-aware structural prior that can differentiate boundary regions from homogeneous interiors, rather than relying on uniform smoothness assumptions. Inspired by recent advancements in edge-guided perceptual networks [47], multibranch mutual-guiding learning [48], and spatial–frequency joint architectures [49] for remote sensing target analysis, which successfully utilize high-frequency (HF) edge components as an explicit structural guide to preserve boundary shapes and maintain structural integrity under severe clutter, we seek to bridge the spatial and frequency domains in HU. To avoid an intuition-driven motivation, we formulate a testable hypothesis to establish a boundary/transition prior for HU. If an HF response extracted from the observed HSI consistently concentrates on material boundaries, it can serve as an effective prior to mitigate severe mixing and error diffusion. On synthetic HSIs with abundance ground truth, we construct an abundance-boundary strength map using the gradient magnitude $\|\nabla \mathbf{A}\|$ and define the boundary mask by thresholding the top 10% responses [see Fig. 1(b)].

The top-10% threshold for constructing the structural boundary mask is chosen as a conservative empirical baseline. This setting is intended to isolate the most prominent spatial gradient transitions across the hyperspectral imagery while strictly preventing the inclusion of independent random noise or minor textures from flat homogeneous interior regions. As shown in Fig. 1(c) and (d), the HF energy obtained by spatially separable 1-D stationary wavelet transform (SS-1D-SWT) exhibits strong spatial correspondence with the ground-truth (GT) boundaries. The quantitative evidence in Table I validates this boundary/transition prior. SWT-based decompositions achieve consistently better boundary alignment (Spearman correlation, ROC-AUC, PR-AUC, and top-10% precision/recall) than 1-D-discrete wavelet transform (DWT) and FFT-based high-pass filtering (FFT-HP).

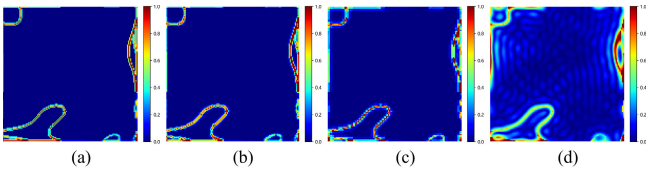


Fig. 2. Visual comparison of HF boundary priors extracted by different decompositions (SS-1D-SWT, 2D-SWT, 1D-DWT, and FFT-HP). Given the same real hyperspectral cube, we compute HF response maps using (a) SS-1D-SWT, (b) 2D-SWT, (c) 1D-DWT, and (d) FFT-HP.

For real HSIs where abundance ground truth is unavailable, we adopt a structural boundary proxy for consistent prior evaluation. We compute the Sobel gradient on the first PCA component (PCA-1) and again threshold the top 10% responses as the boundary mask. This setting is kept consistent with the synthetic study to ensure the robustness and stability of the comparative analysis. Results on Samson and Houston (Table II) show that SWT-based priors (1-D/2-D) provide stronger boundary alignment, whereas FFT-HP suffers from ringing and widespread spurious activations, and DWT tends to fragment boundary responses due to downsampling-induced information loss and subband misalignment. Fig. 2 further visualizes these differences. SWT-derived HF maps yield more concentrated and spatially coherent activations around material boundaries and transition zones, while 1D-DWT weakens/fragments edges and FFT-HP

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introduces nonboundary artifacts. Considering the tradeoff among alignment quality, robustness, and efficiency, we adopt SS-1D-SWT as an efficient implementation to extract the validated HF boundary prior. Its shift invariance and absence of downsampling prevent subband misalignment while achieving near-best boundary alignment with favorable computational cost across datasets (Table II). It is crucial to qualify that the boundary alignment analysis on real datasets relies on the Sobel gradient of the first PCA component as an indicative structural proxy. While this proxy effectively captures major spatial transitions when explicit GT abundance maps are unavailable, it remains inherently susceptible to nonboundary variations such as localized illumination shading and fine surface textures. Consequently, the performance trends in Table II should be interpreted as a relative indicator of spatial alignment capability rather than an absolute measure of material boundary extraction precision.

Building upon the validated boundary/transition prior and its frequency decomposition, we propose a transition-aware boundary-guided dual-frequency network (TABG-Net). The low-frequency (LF) branch models global material distributions and structural consistency, whereas the HF branch captures boundary-related variations and fine structural details. We instantiate the same similarity-aware region attention module (SRAM) in both branches to enable branch-specific region enhancement. SRAM reinforces region continuity of dominant materials in the LF branch and strengthens edge-aware responses while suppressing redundant textures in the HF branch. A boundary-guided module then adaptively integrates the two branches under the guidance of the boundary prior derived from the HF responses, enabling more precise modeling of transition regions. In addition, we design a global guidance branch that operates directly on the original HSI. Through multistage cross-domain feature enhancement module (CFEMs), this branch interacts with the dual-frequency branches to compensate for potential information loss introduced by decomposition and to enrich the spectral–spatial contextual representation, improving both boundary robustness and overall fidelity. Specifically, the main contributions of this study are summarized as follows.

1) We formulate and validate a boundary/transition prior for HU. Specifically, we demonstrate that HF responses extracted from the observed HSI consistently highlight material transition boundaries, helping mitigate severe mixing and error diffusion in transition zones.
2) We propose TABG-Net, a transition-aware boundary-guided dual-frequency network that explicitly embeds the validated boundary/transition prior into a frequency-aware unmixing architecture. A 1-D SWT is adopted to decompose the HSI into LF and HF components, where the LF branch captures global material distributions and smooth variations, while the HF branch focuses on boundary-sensitive details. A boundary-guided module adaptively integrates both streams to improve abundance estimation, especially near transitions.
3) To improve representation completeness and robustness, we introduce a global guidance pathway that operates on the original HSI and interacts with both frequency branches via CFEMs to compensate for potential information loss introduced by decomposition and embed a SRAM into both branches to enhance intraregion consistency and stabilize boundary-relevant responses.

The subsequent structure is outlined as follows. Section II delves into pertinent prior work. Section III presents the proposed TABG-Net. Section IV reports the experimental outcomes for the simulated and real datasets. Finally, the overarching conclusion is summarized in Section V.

II. Related Works

In this section, we briefly review the major research paradigms that underpin HU. Specifically, we discuss: 1) traditional linear unmixing models; 2) DL-based extensions built upon the AE framework; and 3) SS-1D-SWT for HF boundary cues, which collectively form the foundation of the proposed TABG-Net.

A. Traditional and Linear Unmixing Models

The LMM serves as the fundamental framework for HU, where each observed pixel is assumed to be a linear combination of several pure spectral signatures. Mathematically, it can be expressed as follows:

$$\mathbf{x}_i = \mathbf{S}\,\mathbf{a}_i + \boldsymbol{\epsilon}_i \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^B$ denotes the observed spectrum of the ith pixel with B spectral bands, $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_M] \in \mathbb{R}^{B\times M}$ is the endmember matrix containing M pure material spectra, $\mathbf{a}_i \in \mathbb{R}^M$ represents the abundance vector associated with pixel i, and $\boldsymbol{\epsilon}_i$ is the residual noise.

To ensure physical interpretability, the following abundance constraints are typically imposed:

$$a_{i,m} \geq 0, \quad \sum_{m=1}^{M} a_{i,m} = 1 \tag{2}$$

where $a_{i,m}$ denotes the proportion of the mth material in pixel i. The first condition enforces nonnegativity, while the second enforces the abundance sum-to-one constraint (ASC), reflecting the fractional composition of mixed pixels in the real world.

Traditional optimization-based methods, such as fully constrained least squares (FCLSs) [50] and NMF [21], have been extensively applied to estimate both endmembers and abundances under these physical constraints. While effective for simple mixtures, these approaches are limited by the strict linearity assumption, which cannot fully describe nonlinear interactions or account for spectral distortions caused by illumination changes, multiple scatterings, and subpixel heterogeneity. Consequently, their accuracy tends to degrade in highly mixed or boundary-dense regions, where nonlinear and spatial effects are more pronounced.

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B. DL-Based HU

DL has recently played a significant role in the field of remote sensing [51], [52], [53]. In HU, it has emerged as a powerful paradigm because of its ability to learn complex nonlinear relationships between endmembers and mixed pixels. Moreover, DL methods demonstrate a remarkable capacity for joint spectral–spatial modeling, enabling the simultaneous extraction of rich spectral features and intricate spatial contextual dependencies [54], [55]. Among these models, AE architectures are particularly popular for their capability to jointly estimate endmembers and abundances in an unsupervised manner.

An AE-based unmixing framework typically consists of an encoder and decoder. The encoder projects the input hyperspectral pixel $\mathbf{x}_i$ into a lower dimensional representation that corresponds to abundance estimation

$$\mathbf{a}_i = f_{\text{enc}}\left(\mathbf{x}_i;\, \boldsymbol{\Theta}_{\text{enc}}\right) \tag{3}$$

where $f_{\text{enc}}(\cdot)$ denotes the encoder mapping parameterized by $\boldsymbol{\Theta}_{\text{enc}}$. The decoder reconstructs the pixel spectrum from the estimated abundances as follows:

$$\hat{\mathbf{x}}_i = f_{\text{dec}}\left(\mathbf{a}_i;\, \boldsymbol{\Theta}_{\text{dec}}\right) = \mathbf{S}\,\mathbf{a}_i \tag{4}$$

where $\mathbf{S}$ is the endmember matrix and $\boldsymbol{\Theta}_{\text{dec}}$ represents the decoder parameters. To satisfy the physical constraints of abundance, the output of the encoder is often normalized through a *softmax* operation

$$a_{i,m} = \frac{\exp\left(z_{i,m}\right)}{\sum_{k=1}^{M}\exp\left(z_{i,k}\right)} \tag{5}$$

where $\mathbf{z}_i$ is the latent activation vector and $a_{i,m}$ denotes the normalized abundance of the mth material.

Recent AE-based networks have introduced additional mechanisms to enhance spectral–spatial representation, such as residual connections, attention modules, and multiscale feature fusion. These architectural enhancements improve reconstruction fidelity and robustness under spectral variability, particularly in complex or noisy environments. Nevertheless, most deep unmixing models still prioritize holistic spectral reconstruction, whereas fine structural details and boundary-related cues remain insufficiently modeled. This observation highlights the continuing need for approaches capable of preserving local structural information and capturing subtle spectral transitions within mixed regions.

C. SS-1D-SWT for HF Boundary Cues

Frequency-domain representations provide a complementary perspective for hyperspectral analysis by decomposing signals into components that highlight either structural trends or sharp variations. Wavelet transforms (WTs) are extensively utilized in hyperspectral tasks—such as denoising and feature extraction—due to their capability to provide a multiscale separation of coarse and fine-scale information.

While spatial 2D-WT is commonly used to capture textures, it often introduces undesired ringing artifacts and incurs significant computational overhead when applied band-wise to high-dimensional HSIs. Furthermore, downsampling in DWTs typically leads to subband misalignment, which is detrimental to constructing a stable boundary prior. To address these issues, we employ the SS-1D-SWT. The SS-1D-SWT avoids downsampling, maintains full spatial resolution, and ensures shift invariance, making it a superior candidate for extracting reliable HF cues.

In our implementation, we adopt the biorthogonal wavelet bior3.3 as the mother wavelet. The choice of this specific basis is motivated by its biorthogonal properties and is further justified by the ablation experiments in Section IV-D, which demonstrate that bior3.3 consistently provides better performance in preserving signal integrity and capturing fine structural details in challenging environments. To maintain localized edge sensitivity and prevent oversmoothing of fine-grained structures, a single-level ($L = 1$) decomposition is performed. Additionally, a symmetric boundary extension strategy is applied during the transform to suppress potential artifacts at the image edges.

For each spectral band, we perform 1D-SWT independently along the horizontal (x) and vertical (y) axes to capture directional sharp variations. Formally, given a hyperspectral cube $\mathbf{X}$ and its bth band image $\mathbf{X}^{(b)}$, the HF detail coefficients are extracted as follows:

$$\begin{aligned}\mathbf{D}_x^{(b)} &= \mathcal{F}_{\text{HF}}\left(\text{SWT}_x\left(\mathbf{X}^{(b)};\mathbf{h}_{\text{low}},\mathbf{h}_{\text{high}}\right)\right)\\ \mathbf{D}_y^{(b)} &= \mathcal{F}_{\text{HF}}\left(\text{SWT}_y\left(\mathbf{X}^{(b)};\mathbf{h}_{\text{low}},\mathbf{h}_{\text{high}}\right)\right)\end{aligned} \tag{6}$$

where $\mathbf{h}_{\text{low}}$ and $\mathbf{h}_{\text{high}}$ denote the low- and high-pass filters of the bior3.3 wavelet, respectively, and $\mathcal{F}_{\text{HF}}(\cdot)$ denotes the operator that selects HF detail coefficients. The per-band spatial HF response $\mathbf{H}^{(b)}$ is then fused by aggregating directional absolute energies

$$\mathbf{H}^{(b)} = \left|\mathbf{D}_x^{(b)}\right| + \left|\mathbf{D}_y^{(b)}\right| \tag{7}$$

and the final HF boundary prior $\mathbf{M}_{\text{HF}}$ is obtained by averaging these responses across all B bands

$$\mathbf{M}_{\text{HF}} = \frac{1}{B}\sum_{b=1}^{B}\mathbf{H}^{(b)}. \tag{8}$$

Intuitively, $\mathbf{M}_{\text{HF}}$ concentrates on spatial locations with significant spectral–spatial variations, which typically coincide with material transition zones. Compared to standard 2D-WT or FFT-based filters, the SS-1D-SWT prior yields more spatially coherent and artifact-free activations, providing a robust structural guide for the subsequent dual-frequency fusion in TABG-Net.

In addition, as quantitatively validated by the alignment metrics in Tables I and II, although 1-D separable operations theoretically risk directional sensitivity or localized aliasing, the proposed SS-1D-SWT achieves an optimal Spearman correlation of 0.8077 and suppresses the median runtime to 28 ms. This demonstrates that the single-level configuration delivers a superior empirical balance between boundary fidelity and computational efficiency, while any minor directional artifacts or weak boundary information loss are successfully compensated and regularized by the downstream learnable edge reinforcement convolution (LERC) module.

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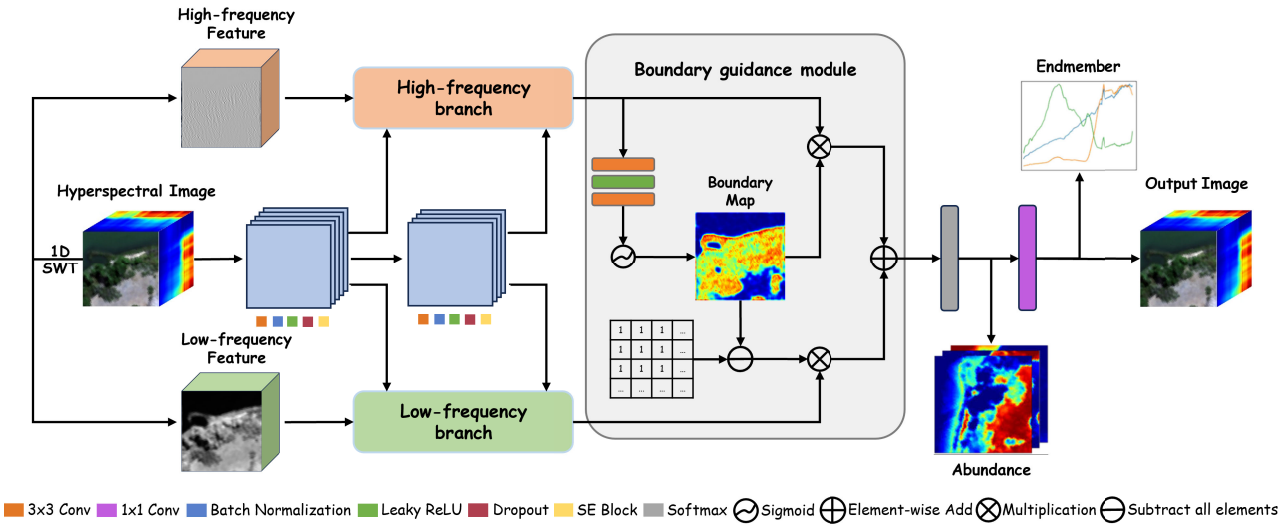


Fig. 3. Architecture of the proposed TABG-Net. The proposed network consists of two complementary branches: the LF branch is designed to model the global spectral structure of dominant materials, while the HF branch focuses on capturing fine-grained textures and edge information.

III. Proposed Method

Fig. 3 illustrates the overall architecture of the proposed TABG-Net. The core philosophy of this design is to separate HU into two specialized frequency-aware tasks: global spectral stabilization and localized boundary preservation. This approach is motivated by the distinct physical characteristics of HSI, aiming to resolve the "smoothing-boundary conflict" inherent in traditional unmixing paradigms. In this context, homogeneous material interiors require spatial consistency to mitigate noise, while transition zones demand HF reinforcement to prevent oversmoothing.

Initially, the input HSI is decomposed by the SS-1D-SWT into LF and HF components. Due to the shift invariance and the absence of downsampling in the wavelet decomposition, these two components remain pixel-wise aligned with the original spatial domain, ensuring consistent feature representation across dual branches. The LF branch utilizes the SRAM to identify and reinforce spectrally similar regions via learnable local difference estimation, focusing on modeling the smooth characteristics of dominant materials to suppress stochastic noise. In contrast, the HF branch functions as a "structure sentinel," employing the LERC module to explicitly amplify sharp structural gradients and capture fine-grained textures that define material interfaces. Additionally, the original HSI is introduced as an auxiliary input to guide spectral–spatial enhancement in the shallow layers, ensuring that the decomposed features remain grounded in the original spectral context.

Finally, the features extracted from both branches are integrated through boundary-guided. By adaptively merging the stabilized spectral signatures from the LF branch with the reinforced structural anchors from the HF branch, TABG-Net produces high-fidelity abundance maps and reconstructed HSIs, denoted as $\hat{\mathbf{X}}$.

Specifically, the predicted abundance fractional maps $\mathbf{A} \in \mathbb{R}^{M\times N}$ are routed to the decoder to complete this spectral reconstruction phase. It is vital to emphasize that the decoder architecture strictly adheres to the standard physical assumption of the LMM. To prevent nonlinear corruptions or structural scaling shifts, the decoder is constructed using a single linear fully connected layer without any nonlinear activation functions or bias parameters, executing the strict linear transformation $\hat{\mathbf{X}} = \mathbf{EA}$, where $\mathbf{E} \in \mathbb{R}^{B\times M}$ represents the endmember signature library. Prior to training, the weights of this linear layer are initialized using vertex component analysis (VCA) to grant a robust physical hot-start. During the optimization stage, the endmember matrix $\mathbf{E}$ is dynamically updated via end-to-end backpropagation, where gradients propagated from the loss objective continuously joint-optimize both the abundance feature extraction branches and the spectral dictionary weights until numerical convergence.

Algorithm 1 summarizes the overall inference pipeline of the proposed method.

A. LF Branch

The LF branch (as shown in Fig. 4) is designed to extract structural features from the LF components obtained via SS-1D-SWT of the original hyperspectral image. These LF components contain global spectral signatures corresponding to the principal materials in the scene. They are characterized by spatial smoothness and structural continuity and are less susceptible to noise owing to their inherent denoising property [56], making them suitable as a foundational representation for HU.

Although LF responses are relatively smooth, they still exhibit weak boundaries and regional transitions. To extract meaningful spatial features from these components, we employed three sequential 3×3 convolutional layers. Each convolutional operation was followed by a LeakyReLU activation, batch normalization (BN), and dropout. LeakyReLU is preferred over ReLU to preserve small negative responses [57], which are crucial for modeling gentle variations in LF signals. BN stabilizes training and mitigates gradient vanishing

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Algorithm 1 Transition-Aware Boundary-Guided Dual-Frequency HU (TABG-Net)

Input: hyperspectral image $\mathbf{X}$
Output: Abundance maps $\mathbf{A}$ and reconstructed image $\hat{\mathbf{X}}$

1: Decompose $\mathbf{X}$ into low- and high-frequency components $(\mathbf{X}_{\text{low}}, \mathbf{X}_{\text{high}})$ via SS-1D-SWT;
2: Extract low-frequency features $\mathbf{F}_{\text{low}}$ for global material modeling from $\mathbf{X}_{\text{low}}$;
3: Extract high-frequency features $\mathbf{F}_{\text{high}}$ for boundary-sensitive representation from $\mathbf{X}_{\text{high}}$;
4: Extract global guidance features $\mathbf{F}_{\text{ori}}$ from the original HSI $\mathbf{X}$;
5: Enhance $\mathbf{F}_{\text{low}}$ and $\mathbf{F}_{\text{high}}$ using cross-frequency enhancement with $\mathbf{F}_{\text{ori}}$;
6: Predict a boundary/transition guidance map $\mathbf{M}$ from high-frequency features $\mathbf{F}_{\text{high}}$;
7: Fuse dual-frequency features using boundary-guided:

$$\mathbf{F} = \mathbf{M} \odot \mathbf{F}_{\text{high}} + (1 - \mathbf{M}) \odot \mathbf{F}_{\text{low}}$$

8: Predict abundance maps $\mathbf{A}$ from fused features;
9: Reconstruct hyperspectral image $\hat{\mathbf{X}}$ from $\mathbf{A}$;
10: **return** $\mathbf{A}, \hat{\mathbf{X}}$

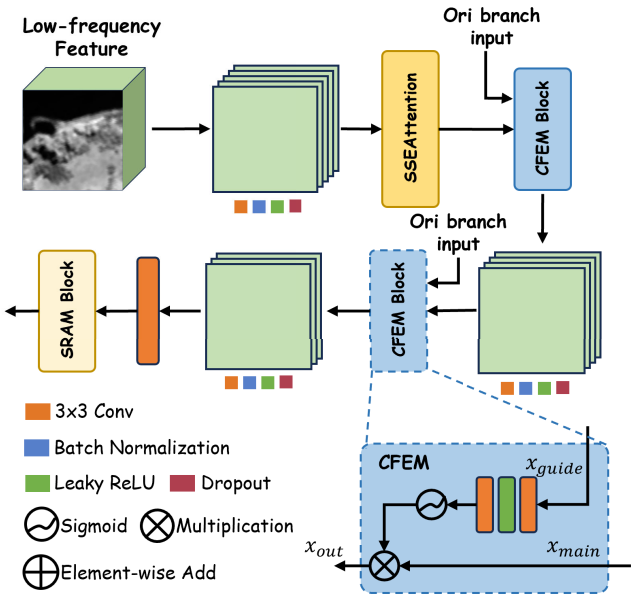


Fig. 4. Architecture of the LF branch. The SSEA module captures global spectral dependencies and local discriminative patterns by combining SE and spectral attention mechanisms. The CFEM modules perform cross-domain feature encoding by modulating LF features with original HSI guidance. The SRAM module further highlights smooth and semantically consistent regions to improve endmember estimation.

[58], whereas dropout helps prevent overfitting by randomly deactivating a subset of feature channels during training [59].

To further bolster the spectral–spatial representation, we introduce the spatial–spectral enhancement attention (SSEA) module. This module sequentially incorporates a squeeze-and-excitation (SE) block and an enhanced spectral attention (ESA) module. The SE block models the nonlinear interdependencies between channels to generate a recalibration weight vector, which is then used to rescale the original features

$$\mathbf{F}_{se} = \sigma\left(W_2 \cdot \delta\left(W_1 \cdot \text{GAP}\left(\mathbf{F}\right)\right)\right) \otimes \mathbf{F} \tag{9}$$

where $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$ denotes the input feature, GAP(·) represents the global average pooling, W_1 and W_2 are 1×1 convolutional layers, $\delta(\cdot)$ is the ReLU activation function, $\sigma(\cdot)$ is the sigmoid function, and $\otimes$ denotes element-wise multiplication.

Subsequently, the ESA module performs a nonlinear mapping on the intermediate feature $\mathbf{F}_{se}$ along the spectral dimension to capture both local correlations and long-range dependencies between bands. This process generates a spectral attention mask $\mathcal{M}_{esa}$ with frequency-selective properties

$$\mathcal{M}_{esa}\left(\mathbf{F}_{se}\right) = \sigma\left(W_u \cdot \delta\left(W_d \cdot \text{GAP}\left(\mathbf{F}_{se}\right)\right)\right). \tag{10}$$

The generated mask is applied to the intermediate features via element-wise multiplication to achieve precise enhancement of specific frequency bands

$$\mathbf{F}_{out} = \mathcal{M}_{esa}\left(\mathbf{F}_{se}\right) \otimes \mathbf{F}_{se} \tag{11}$$

where W_d and W_u represent the learnable parameter matrices for dimensionality reduction and restoration, respectively, within the spectral attention branch.

To address the underexpression of LF content and enrich global perception, we leverage the original HSI as a supplementary guide via a guided feature modulation (CFEM) module. This module modulates the LF features with guidance from the original HSI

$$\mathbf{F}_{\text{guide}} = \text{Conv}^{(2)}_{1\times1}\left(\delta\left(\text{Conv}^{(1)}_{1\times1}\left(\mathbf{F}_{\text{ori}}\right)\right)\right) \tag{12}$$

$$\mathbf{F}_{\text{enhanced}} = \mathbf{F}_{\text{low}} \odot \sigma\left(\mathbf{F}_{\text{guide}}\right) \tag{13}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{F}_{\text{low}}$ is the current LF feature map.

The CFEM branch functions strictly as a shallow spectral anchor in the initial layer of the network. Rather than performing complex nonlinear feature matching or deep representations, it employs lightweight 1×1 convolutions for basic spectral–spatial guidance. Because the original HSI and the dual-frequency components are inherently pixel-wise aligned due to the properties of the SS-1D-SWT, explicit spatial alignment mechanisms are structurally redundant. This minimalist design ensures that the decoupled features remain physically grounded in the original spectral context during early transformation stages, avoiding the risk of feature dilution or overparameterization in deeper layers.

Finally, to suppress interference from irrelevant textures and noise in the LF context, we introduce the SRAM. This module identifies and reinforces features within homogeneous regions to enhance abundance smoothness while filtering out minor structural clutter.

Unlike traditional fixed-threshold masks, the identification of similar regions in SRAM is driven by a learnable local difference estimation. Specifically, the convolutional weights W_1 and W_2 are optimized to act as adaptive variance estimators. In regions where spectral–spatial signatures are consistent, the resulting difference map $\mathbf{D}$ yields low response values, signifying high pixel-to-pixel similarity.

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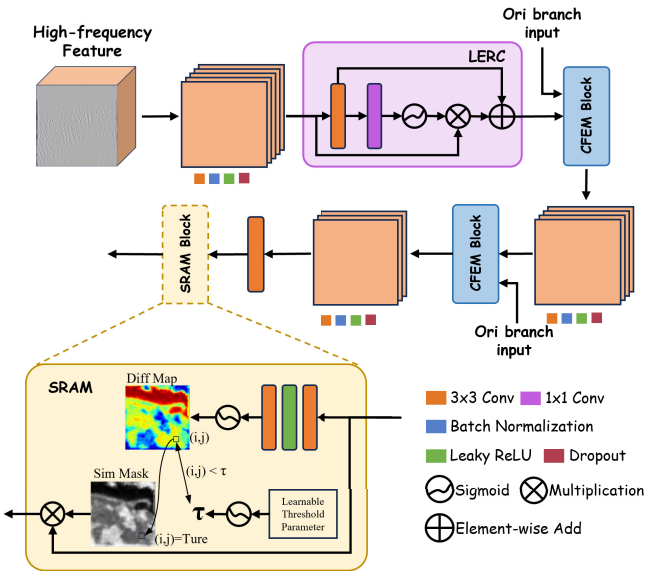


Fig. 5. Architecture of the HF branch. The LERC module is designed to enhance boundary-sensitive representations by learning edge-aware features. The SRAM module emphasizes structurally similar regions by enhancing locally coherent textures for improved spectral–spatial consistency.

To ensure structural edge preservation and accurately capture the subtle material transitions common in hyperspectral imagery, SRAM employs a rigid binary similarity gate M to prevent spatial oversmoothing. The process is formulated as follows:

$$\mathbf{D} = \sigma\left(W_2 * \delta\left(W_1 * \mathbf{F}\right)\right) \tag{14}$$

$$\mathbf{M} = \mathbf{1}_{(\mathbf{D}<\sigma(\tau))} \tag{15}$$

$$\mathbf{F}' = \mathbf{F} \odot (1 + \mathbf{M}) \tag{16}$$

where W_1 and W_2 denote convolutional weights, τ is a learnable similarity threshold, and $\sigma(\cdot)$ represents the sigmoid activation function, which constrains both the difference map D and the threshold to $[0, 1]$. The term $\mathbf{1}_{(\cdot)}$ acts as an indicator function that generates a hard binary routing mask. This gating mechanism allows TABG-Net to strictly restrict feature reinforcement ($\mathbf{F} \odot 2$) within highly consistent, homogeneous interior zones, while executing an identity mapping ($\mathbf{F} \odot 1$) for boundary pixels to rigidly preserve original HF spatial gradients and structural sharp edges.

B. HF Branch

The HF branch (as shown in Fig. 5) is designed to extract spatial detail and boundary information from the HF components derived through the SS-1D-SWT of the original hyperspectral image. These components highlight abrupt spatial variations, such as object edges and fine textures, making them essential for improving the spatial precision of unmixing results [60]. However, HF signals are inherently more sensitive to noise, requiring targeted enhancement and regulation strategies.

To retain sharp spatial transitions and enhance discriminative details, the HF branch uses three 3×3 convolutional blocks. Although this structure is similar to that of the LF path, its role emphasizes fine-grained localization and boundary extraction rather than structural smoothing. Each block adopts LeakyReLU, BN, and dropout to preserve edge sensitivity and ensure robust training under noisy conditions.

To further strengthen boundary-sensitive representations, we introduce an LERC module. It is worth noting that while the input to this branch consists of frequency-decomposed HF components, these components maintain full spatial resolution and shift invariance due to the properties of SS-1D-SWT. Consequently, the HF subbands can be interpreted as spatial detail-energy maps that are pixel-wise aligned with the original hyperspectral cube $\mathbf{X}$. This alignment ensures that spatial-domain gradient operators can be effectively applied to further analyze the localized HF energy.

The LERC module leverages this interpretability by applying a depthwise edge-detection convolution layer EdgeConv$(\cdot)$, which acts as a spatial gradient operator to achieve second-order edge enhancement. To provide a structural inductive bias, the kernels within EdgeConv$(\cdot)$ are initialized with Sobel-like filters. It should be emphasized that these kernels are learnable parameters rather than fixed operators. This initialization serves as a physically meaningful hot-start, guiding the network to prioritize physical material transitions from the beginning of training, while allowing the kernels to adaptively refine based on the specific spatial–spectral nuances of the scene. The LERC consists of this edge-detection layer and a gating mechanism that amplifies informative edge structures while suppressing noise

$$\mathbf{F}_{\text{high}} = \text{EdgeConv}\left(\mathbf{F}\right) \tag{17}$$

$$\mathbf{A}_{\text{edge}} = \sigma\left(\text{Gate}\left(\mathbf{F}_{\text{high}}\right)\right) \tag{18}$$

$$\mathbf{F}' = \mathbf{F} \odot \left(1 + \mathbf{A}_{\text{edge}}\right) + \mathbf{F}_{\text{high}} \tag{19}$$

where $\mathbf{F}$ is the input HF feature, σ denotes the sigmoid activation, and Gate$(\cdot)$ is a 1×1 convolution that generates the attention map $\mathbf{A}_{\text{edge}}$. By grounding the HF branch in learnable physical gradients, this design ensures that TABG-Net can effectively distinguish sharp material transitions from stochastic noise, resulting in superior boundary fidelity.

It is vital to clarify that the LERC module does not functionally duplicate the SS-1D-SWT, but rather establishes a hierarchical refinement paradigm. While the SS-1D-SWT extracts raw, unselective HF detail-energy maps that inherently contain both true boundaries and stochastic noise, the LERC module utilizes these maps as its structural inputs. By initializing the edge-detection convolution with Sobel-like kernels as a structural inductive bias, the network is granted a physically meaningful hot-start. Since the kernels are fully learnable parameters rather than fixed operators, they adaptively optimize their weights during end-to-end training to attenuate random noise variations and avoid gradient instability under low SNR conditions, thereby isolating robust and continuous material interfaces.

To provide HF features with global semantic context and suppress redundant texture responses, we inject guidance from the original hyperspectral image using CFEMs. These modules operate in the first two stages to modulate current features

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using original HSI projections

$$\mathbf{F}_{\text{guide}} = \text{Conv}^{(2)}_{1\times1}\left(\delta\left(\text{Conv}^{(1)}_{1\times1}\left(\mathbf{F}_{\text{ori}}\right)\right)\right) \tag{20}$$

$$\mathbf{F}_{\text{enhanced}} = \mathbf{F}_{\text{high}} \odot \sigma\left(\mathbf{F}_{\text{guide}}\right) \tag{21}$$

where $\mathbf{F}_{\text{high}}$ is the intermediate HF feature map and $\delta(\cdot)$ denotes the LeakyReLU activation. This mechanism helps suppress irrelevant noise responses and maintain a meaningful structure by aligning HF features with global semantic priors from the original input.

Finally, a SRAM is employed to reinforce spatial coherence within textural regions. Unlike its application in the LF branch, where SRAM enhances homogeneous areas for smoother abundance maps, it focuses on highlighting structurally consistent edge patterns. By adaptively emphasizing semantically similar fine structures, SRAM suppresses HF noise while preserving the details essential for accurate unmixing.

C. Original HSI Guidance Branch

To enhance the model's capacity for global spectral-context modeling, we introduced the original hyperspectral image (HSI) as a third auxiliary guidance branch. Unlike the LF and HF branches, the original HSI branch does not directly contribute to the final unmixing outputs. Instead, it interacts with the main LF and HF pathways in a stage-wise manner, providing cross-branch guidance to promote more comprehensive spectral and spatial representations.

The original HSI input was first processed through two convolutional blocks to extract semantic guidance features. Each block consists of a 3×3 convolution followed by LeakyReLU activation, BN, and an SE module, which compresses the channel dimension and models global dependencies to generate compact contextual features [61]. This extraction process can be formally expressed as follows:

$$\mathbf{F}^{(1)}_{\text{ori}} = \text{SE}\left(\text{BN}\left(\delta\left(\text{Conv}_{3\times3}\left(\mathbf{X}\right)\right)\right)\right) \tag{22}$$

$$\mathbf{F}^{(2)}_{\text{ori}} = \text{SE}\left(\text{BN}\left(\delta\left(\text{Conv}_{3\times3}\left(\mathbf{F}^{(1)}_{\text{ori}}\right)\right)\right)\right) \tag{23}$$

where $\mathbf{X}$ is the input original hyperspectral image, $\delta(\cdot)$ denotes the LeakyReLU activation, BN$(\cdot)$ represents BN, and SE$(\cdot)$ denotes the SE operation.

The resulting feature $\mathbf{F}_{\text{ori}}$ is then used to guide the main pathways via the cross-domain feature enhancement module (CFEM). This is achieved by modulating the intermediate features of the LF and HF branches using a gated attention mechanism derived from the original HSI context

$$\mathbf{F}'_{\text{low}} = \mathbf{F}_{\text{low}} \odot \sigma\left(\text{Conv}^{(2)}_{1\times1}\left(\delta\left(\text{Conv}^{(1)}_{1\times1}\left(\mathbf{F}_{\text{ori}}\right)\right)\right)\right) \tag{24}$$

$$\mathbf{F}'_{\text{high}} = \mathbf{F}_{\text{high}} \odot \sigma\left(\text{Conv}^{(2)}_{1\times1}\left(\delta\left(\text{Conv}^{(1)}_{1\times1}\left(\mathbf{F}_{\text{ori}}\right)\right)\right)\right) \tag{25}$$

where $\odot$ denotes element-wise multiplication and $\sigma(\cdot)$ is the sigmoid activation. Through feature-wise modulation, the original HSI context acts as a global spectral prior, enabling the main branches to adaptively focus on structurally and spectrally salient regions, thus improving the representation quality and unmixing accuracy.

D. Boundary-Guided Fusion Mechanism

To integrate complementary information from LF and HF branches more effectively, we propose a boundary-guided mechanism that dynamically adjusts the reliance on different branches across spatial regions. The HF branch captures rich spatial detail and edge information, while the LF branch focuses on structural coherence and dominant material representation. Thus, we introduce a dedicated boundary extractor module to generate a boundary response map that guides the final feature fusion.

Specifically, the boundary map is derived from the deep features of the HF branch. The feature map from the third stage, denoted as $\mathbf{F}^{(3)}_{\text{high}}$, is passed through two stacked 3×3 convolutional layers, followed by ReLU and sigmoid activations to yield a boundary probability map

$$\mathbf{M}_{\text{edge}} = \sigma\left(\text{Conv}^{(2)}_{3\times3}\left(\delta\left(\text{Conv}^{(1)}_{3\times3}\left(\mathbf{F}^{(3)}_{\text{high}}\right)\right)\right)\right) \tag{26}$$

where $\delta(\cdot)$ denotes the ReLU activation function and $\sigma(\cdot)$ is the sigmoid function. The resulting map $\mathbf{M}_{\text{edge}}$ highlights pixels likely to belong to edge regions.

The final fused feature map is then obtained by modulating the contributions of the LF and HF final features using the boundary map

$$\mathbf{F}_{\text{fused}} = \mathbf{M}_{\text{edge}} \cdot \mathbf{F}^{\text{final}}_{\text{high}} + \left(1 - \mathbf{M}_{\text{edge}}\right) \cdot \mathbf{F}^{\text{final}}_{\text{low}}. \tag{27}$$

This adaptive fusion strategy ensures that HF details are emphasized in boundary regions, whereas LF smoothness is preserved in nonboundary areas. Consequently, the fused representation maintains structural consistency and produces more accurate and visually sharp unmixing outputs.

It should be noted that the predicted boundary-guidance map $\mathbf{M}_{\text{edge}}$ is optimized without explicit boundary supervision. This design choice is motivated by the inherent difficulty of obtaining precise boundary ground truth in remote sensing. However, the risk of unconstrained self-reinforcement is mitigated by the frequency-domain decoupling strategy.

Since the input to the HF branch is strictly constrained to the detail-energy subbands extracted by SS-1D-SWT, the gating mechanism can only reweight and refine the existing spatial-detail energy. Furthermore, the reconstruction loss $\mathcal{L}_{\text{SAD}}$ provides an implicit physical constraint; because pixels in transition zones exhibit higher spectral ambiguity, the network is naturally incentivized to leverage the reinforced HF structural cues to minimize global reconstruction error. This ensures that $\mathbf{M}_{\text{edge}}$ captures true physical material interfaces rather than serving as a purely numerical artifact for reconstruction.

E. Loss Function

In the absence of GT abundance annotations, the proposed network is trained in an unsupervised manner using a spectral reconstruction loss. Specifically, we adopt the spectral angle distance (SAD) to measure the spectral similarity between the reconstructed hyperspectral image and the original input.

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Given the original spectral vector $\mathbf{x}_i \in \mathbb{R}^B$ and the reconstructed spectral vector $\hat{\mathbf{x}}_i \in \mathbb{R}^B$ at the ith pixel, the SAD is defined as follows:

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N} \arccos\left(\frac{\langle \hat{\mathbf{x}}_i, \mathbf{x}_i\rangle}{\|\hat{\mathbf{x}}_i\|_2 \, \|\mathbf{x}_i\|_2}\right) \tag{28}$$

where N denotes the total number of pixels, $\langle\cdot,\cdot\rangle$ represents the inner product, and $\|\cdot\|_2$ denotes the ℓ_2 norm.

It is essential to clarify the rationale behind utilizing the SAD loss as the primary reconstruction constraint instead of the conventional mean squared error (mse). While mse tracks absolute radiometric magnitudes, it exhibits high sensitivity to illumination variations, topographic shading, and sensor scaling noise, which frequently misguides the optimization process and corrupts abundance estimations in real-world environments. In contrast, the scale-invariant property of the SAD loss enables the network to focus exclusively on intrinsic spectral-shape similarities, inherently factoring out localized brightness variations. Furthermore, the absolute magnitude of the abundance estimation is rigorously anchored through the simultaneous enforcement of the abundance nonnegativity constraint (ANC) and ASC in the bottleneck layer. Since the endmember dictionary preserves the physical radiometric scales of materials and the fractional abundances are mathematically bounded to sum to one, minimizing the spectral-shape divergence via the SAD loss is mathematically sufficient and physically more robust to guide the network toward high-precision unmixing performance.

IV. Experiments

In this section, the effectiveness of the proposed TABG-Net is evaluated on a synthetic dataset and three real hyperspectral datasets. To validate its performance, we compare it with seven representative and competitive unmixing methods, including both classical and DL-based approaches: VCA + FCLS [50], [62], DAEN [27], CyCU-Net [63], DeepTrans [64], A2SAN [65], SWC-Net [66], and SRViT [67].

It is important to note that traditional standalone frequency-domain or wavelet-domain unmixing methods, which typically operate as isolated preprocessing or thresholding pipelines, are omitted from the global comparison due to structural and optimization paradigm mismatches with end-to-end unsupervised deep AEs. Instead, the selected baseline configuration focuses on state-of-the-art DL architectures that represent the cutting edge of spatial–spectral contextual modeling. In this context, the frequency-domain component of TABG-Net functions as an integrated feature-decoupling mechanism rather than an isolated transform domain, making deep spatial–spectral paradigms the most fair and rigorous benchmarks to demonstrate its superiority in resolving the smoothing-boundary conflict.

A. Dataset Descriptions

We used one synthetic dataset and three real-world datasets, as shown in Fig. 6. The details are provided next.



(a) (b) (c) (d)

Fig. 6. Pseudo-RGB images of the dataset. (a) Synthetic data, (b) Samson, (c) Houston, and (d) Moffett.

1) Synthetic Dataset: The synthetic dataset is constructed using reflectance spectral signatures selected from the United States Geological Survey (USGS) spectral library. Each synthetic hyperspectral image has a spatial resolution of 100×100 pixels and contains 200 spectral bands uniformly sampled over the wavelength range from 0.38 to 2.5 μm. To better align with the motivation of this work, the synthetic scenes are designed to exhibit spatially continuous material distributions with smoothly varying transition zones between adjacent materials, where mixed pixels are more prevalent. Crucially, to ensure a rigorous evaluation and prevent pure-pixel shortcuts, the simulated abundance maps are generated under a strict no-pure-pixel constraint by capping the maximum fractional abundance for any individual endmember within a single pixel at approximately 0.9. This configuration significantly increases the unmixing difficulty, ensuring that all comparative methods and the proposed TABG-Net are evaluated under a highly mixed and realistic scenario. Furthermore, additive Gaussian noise with SNRs of 20, 30, and 40 dB is introduced to evaluate the robustness of different HU methods under varying noise levels.

2) Samson Dataset: The Samson dataset is a widely used real-world dataset for HU research, acquired by the airborne hyperspectral imaging system Spectroscopic Aerial Mapper with Onboard Navigation (SAMSON). The image has a spatial resolution of 95×95 pixels and consists of 156 spectral bands covering wavelengths from 401 to 889 nm. This dataset primarily contains three endmembers: soil, trees, and water.

3) Houston Dataset: The Houston dataset was originally released for the 2013 IEEE GRSS Data Fusion Contest and has since been widely used to test and benchmark HU methods. It has a spatial size of 105×105 pixels and contains 144 spectral bands covering wavelengths from 380 to 1050 nm. The image primarily includes four land cover types: grass, runway, parking lot 1, and parking lot 2.

4) Moffett Dataset: The Moffett dataset was captured by the well-known Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and is considered a challenging benchmark for HU. The image has a spatial size of 50×50 pixels and consists of 184 spectral bands. The main land cover types are water, soil, and vegetation.

B. Experimental Settings

To ensure a fair and comprehensive comparison, all baseline methods included in our study are implemented using their respective publicly available source codes. For deep-learning-based paradigms, a unified training protocol is maintained across all experiments to eliminate potential biases. This

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TABLE III
HYPERPARAMETER TUNING RESULTS ON THE HOUSTON DATASET

Learning Rate	Weight Decay	Epochs	mSAD ↓
Effect of learning rate (WD=5×10^{-4}, Epochs=500)			
0.001	5×10^{-4}	500	0.093
0.004	5×10^{-4}	500	**0.072**
0.01	5×10^{-4}	500	0.083
Effect of weight decay (LR=0.004, Epochs=500)			
0.004	1×10^{-4}	500	0.087
0.004	$\mathbf{5 \times 10^{-4}}$	500	**0.072**
0.004	1×10^{-3}	500	0.077
Effect of training epochs (checkpoints)			
0.004	5×10^{-4}	200	0.097
0.004	5×10^{-4}	400	0.077
0.004	$\mathbf{5 \times 10^{-4}}$	**500**	**0.072**

includes utilizing the Adam optimizer with a consistent maximum epoch limit and an identical early-stopping strategy with a patience of 50. The specific hyperparameters for each baseline, such as the learning rate and dataset-specific regularization coefficients, were meticulously tuned via a comprehensive grid search on each dataset $\mathbf{X}$ to report their optimal performance. This standardized experimental setup guarantees that the performance gains of TABG-Net are attributed to its underlying model architecture rather than variations in training intensity or implementation quality.

1) Hyperparameter Settings: The proposed TABG-Net is implemented using the PyTorch framework and trained on a single NVIDIA GeForce RTX 3060 GPU. Following the standard practice in blind HU, the batch size is set to 1, meaning that the entire hyperspectral cube $\mathbf{X}$ is treated as a single training unit per iteration. This configuration is motivated by the transductive nature of the unmixing task, which focuses on the self-reconstruction and feature extraction of a specific target scene rather than learning a generalized model from a large-scale dataset.

Under this single-batch setting, the stability of the normalization layers is maintained by the large spatial dimensions ($H \times W$) of the HSI. Since the mean and variance are estimated across thousands of pixels within the feature map $\mathbf{F}$, the normalization process remains statistically robust. Furthermore, Dropout is applied with a rate of 0.1 to prevent overfitting of the specific spatial structures. Empirical observations show that this training protocol ensures stable gradient descent and fast convergence across all tested datasets, with no signs of numerical instability typically associated with small batch sizes in inductive learning tasks.

A staged hyperparameter tuning strategy is adopted. The learning rate, weight decay, and number of training epochs are first tuned on a representative real-world dataset (Houston) to ensure stable convergence, as reported in Table III. Specifically, the learning rate and weight decay are selected from small candidate sets, while the number of epochs is determined based on convergence behavior using intermediate checkpoints.

TABLE IV
FINAL TRAINING HYPERPARAMETERS FOR DIFFERENT DATASETS

Dataset	Epochs	LR	Weight Decay
Synthetic	500	1×10^{-3}	1×10^{-4}
Samson	600	3×10^{-2}	2×10^{-4}
Houston	500	4×10^{-3}	5×10^{-4}
Moffett	500	1×10^{-2}	5×10^{-4}

To prevent any optimization bias, the tuning intensity for both TABG-Net and the baseline methods was kept strictly consistent. Specifically, each model underwent an intensive hyperparameter search on the Houston dataset to establish its optimal base configuration. When evaluating performance on other datasets, a uniform transfer protocol was followed. We applied consistent and minor adjustments to the learning rate and regularization terms for all methods to accommodate the specific noise levels and spectral characteristics of each scene. This standardized approach ensures that all compared paradigms, from traditional geometric methods to modern Transformer-based architectures, are evaluated at their respective peak performance levels, thereby maintaining the integrity of the comparative study.

The selected configuration is then applied to the remaining datasets, with only minor adjustments of the learning rate and regularization strength to accommodate different convergence characteristics. The learning rate scheduler parameters are fixed across all datasets, with a step size of 40 and a decay factor of 0.8. All final hyperparameter settings for different datasets are summarized in Table IV.

2) Evaluation Metrics: To evaluate the HU performance, we adopt two widely used quantitative metrics: root mse (RMSE) and SAD.

RMSE measures the reconstruction accuracy of abundance maps by computing the Euclidean distance between the estimated abundances and the ground truth. For the pth endmember over N pixels, it is defined as follows:

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(a_{i,p} - \hat{a}_{i,p}\right)^2} \tag{29}$$

where $a_{i,p}$ and $\hat{a}_{i,p}$ denote the GT and estimated abundances, respectively. The average RMSE over all P endmembers is given by

$$\mathrm{aRMSE} = \frac{1}{P}\sum_{p=1}^{P}\mathrm{RMSE}_p. \tag{30}$$

SAD evaluates the angular discrepancy between the estimated and GT endmember spectra in the spectral space and is defined as follows:

$$\mathrm{SAD}_p = \arccos\left(\frac{\langle \mathbf{e}_p, \hat{\mathbf{e}}_p \rangle}{\|\mathbf{e}_p\|_2 \, \|\hat{\mathbf{e}}_p\|_2}\right) \tag{31}$$

where $\mathbf{e}_p$ and $\hat{\mathbf{e}}_p$ denote the GT and estimated spectra of the pth endmember, respectively. The average SAD is computed as follows:

$$\mathrm{aSAD} = \frac{1}{P}\sum_{p=1}^{P}\mathrm{SAD}_p. \tag{32}$$

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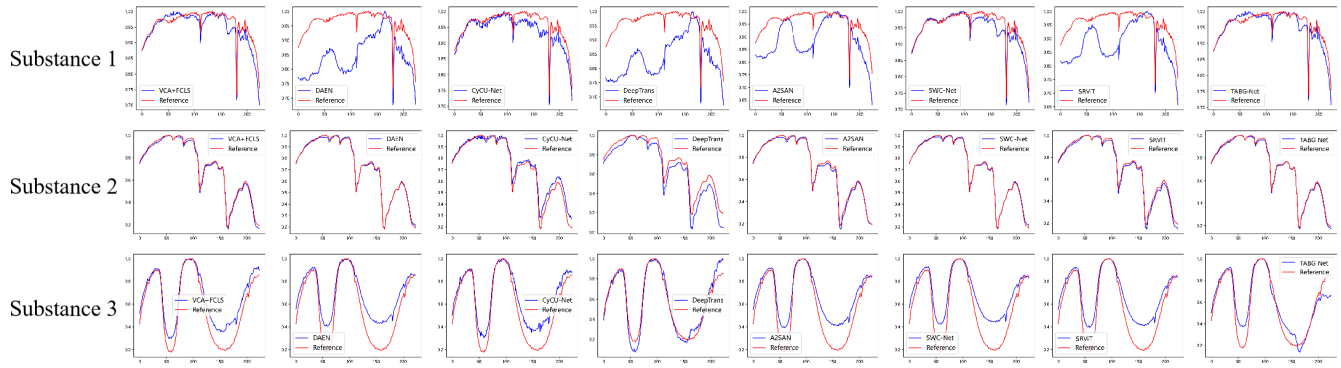


Fig. 7. Visual comparison of the endmembers extracted by different unmixing methods on the synthetic dataset (30 dB).

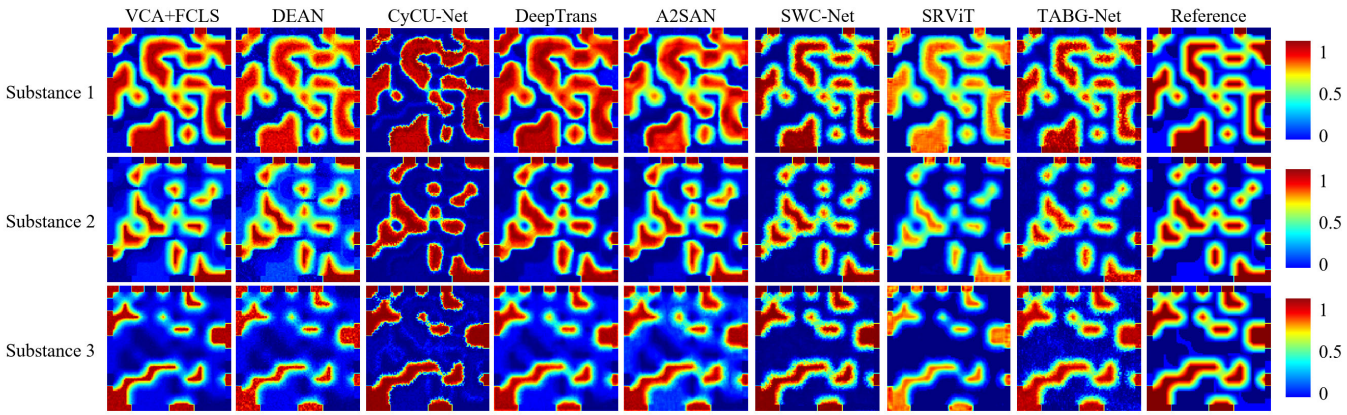


Fig. 8. Visual comparison of the abundance maps estimated by different unmixing methods on the synthetic dataset (30 dB).

TABLE V
COMPARISON OF MODEL COMPLEXITY WITH EXISTING METHODS

Method	GFLOPs ↓	Params (M) ↓
DAEN	1.43	0.16
CyCU-Net	0.77	0.04
DeepTrans	1.00	4.97
A2SAN	0.47	0.05
SWC-Net	10.66	1.25
SRViT	147.08	39.54
TABG-Net	7.29	0.82

In addition to accuracy metrics, the model complexity, in terms of the number of parameters and GFLOPs, is reported to assess the computational efficiency of different methods. To ensure a rigorous and reproducible evaluation, the GFLOPs for all compared methods are profiled using a standardized input tensor with a spatial size of 95×95 pixels and 156 spectral bands, which directly corresponds to the configuration of the Samson dataset. The results are presented in Table V. The proposed TABG-Net exhibits a moderate computational footprint. Specifically, its parameter count (0.82 M) and GFLOPs (7.29) are significantly lower than those of the Vision Transformer-based SRVIT, which requires approximately 48 times more parameters and 20 times more GFLOPs. Although TABG-Net is slightly more complex than basic AE models like DAEN or A2SAN, this additional overhead is primarily attributed to the dual-stream architecture designed for frequency-separable modeling. The experimental results across multiple datasets demonstrate that this increase in complexity is justified by the substantial gains in Mean SAD and Mean RMSE, particularly in preserving boundary details where lighter models typically underperform.

C. Experiments on the Synthetic Dataset

Table VI reports the quantitative results on the synthetic dataset under 20-, 30-, and 40-dB noise levels. TABG-Net consistently achieves the lowest Mean SAD and Mean RMSE across all configurations, demonstrating superior noise robustness. Particularly at the challenging 20-dB level, TABG-Net outperforms standard AEs (DAEN and CyCU-Net) and advanced spatial–spectral or frequency-domain architectures (SRViT and SWC-Net) by a clear margin. To intuitively demonstrate the unmixing capability under no-pure-pixel scenarios, Figs. 7 and 8 provide the qualitative endmember spectral curves and abundance estimations under a representative 30-dB noise level.

This superiority confirms that the proposed boundary-guided dual-frequency modeling successfully decouples global LF structures from HF boundary variations, effectively suppressing error diffusion in transition zones. As the SNR scales up to 40 dB, although cleaner observations narrow the performance gaps, TABG-Net consistently maintains the

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TABLE VI

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE SYNTHETIC DATASET UNDER DIFFERENT NOISE LEVELS. THE BEST RESULTS ARE IN BOLD

SNR	Metric	VCA+FCLS	DAEN	CyCU-Net	DeepTrans	A2SAN	SWC-Net	SRViT	TABG-Net
20 dB	Mean SAD	0.084 ± 0.005	0.119 ± 0.005	0.131 ± 0.040	0.089 ± 0.018	0.240 ± 0.016	0.078 ± 0.004	0.105 ± 0.002	**0.063 ± 0.006**
	Mean RMSE	0.117 ± 0.005	0.158 ± 0.010	0.181 ± 0.006	0.161 ± 0.018	0.333 ± 0.020	0.114 ± 0.005	0.154 ± 0.006	**0.097 ± 0.003**
30 dB	Mean SAD	0.077 ± 0.004	0.115 ± 0.007	0.097 ± 0.010	0.076 ± 0.001	0.113 ± 0.019	0.072 ± 0.002	0.097 ± 0.005	**0.056 ± 0.002**
	Mean RMSE	0.107 ± 0.006	0.130 ± 0.004	0.201 ± 0.011	0.117 ± 0.007	0.170 ± 0.012	0.080 ± 0.005	0.108 ± 0.006	**0.075 ± 0.003**
40 dB	Mean SAD	0.089 ± 0.006	0.101 ± 0.005	0.088 ± 0.008	0.073 ± 0.006	0.092 ± 0.006	0.069 ± 0.003	0.093 ± 0.004	**0.055 ± 0.002**
	Mean RMSE	0.099 ± 0.006	0.140 ± 0.002	0.181 ± 0.007	0.107 ± 0.013	0.155 ± 0.004	0.075 ± 0.003	0.101 ± 0.004	**0.069 ± 0.001**

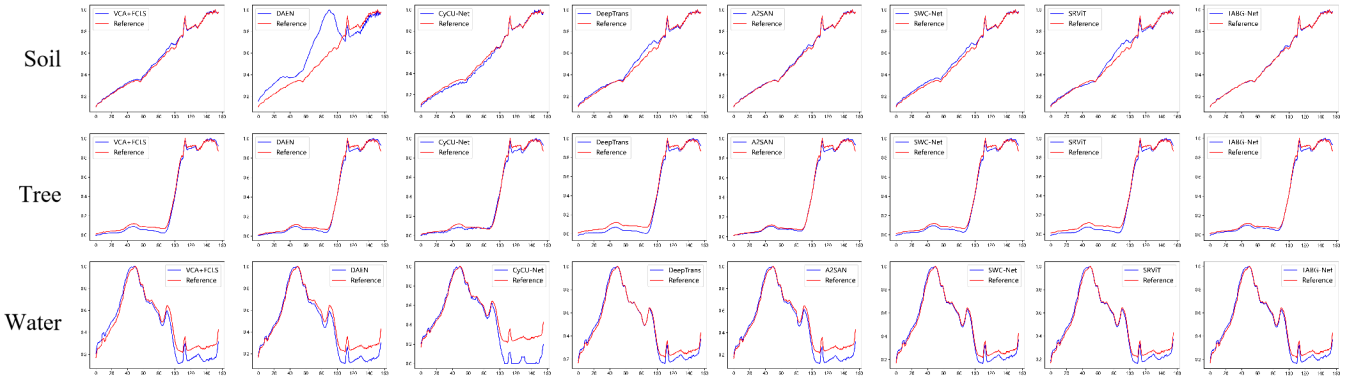


Fig. 9. Visual comparison of the endmembers extracted by different unmixing methods on the Samson dataset.

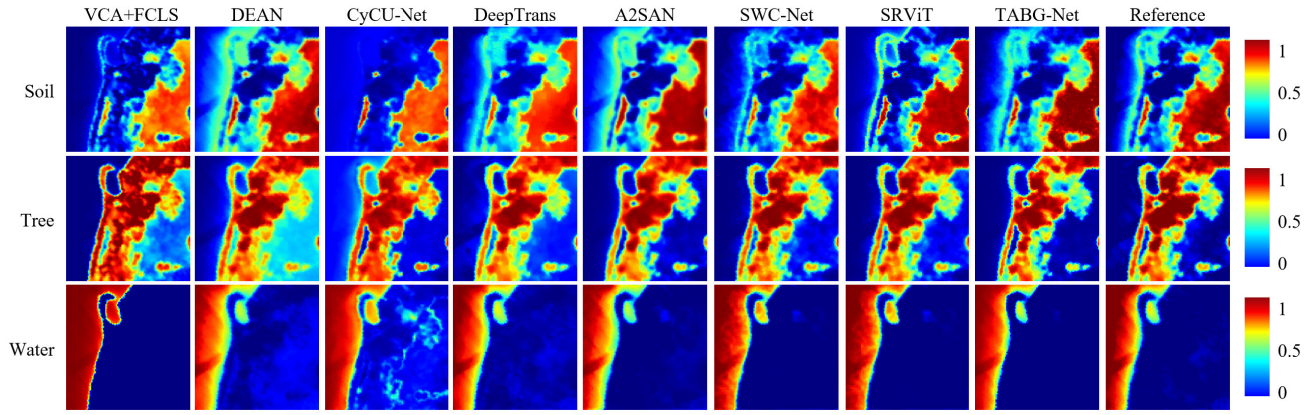


Fig. 10. Visual comparison of the abundance maps estimated by different unmixing methods on the Samson dataset.

optimal performance tier, verifying its stable edge-enforcement capabilities across diverse noise conditions.

D. Experiments on the Samson Dataset

Table VII summarizes the quantitative metrics on the Samson dataset. Overall, TABG-Net delivers the best Mean SAD (0.034) and Mean RMSE (0.040) among all competitors, proving its superior global unmixing performance. At the class level, TABG-Net consistently achieves the lowest RMSE for soil, tree, and water, demonstrating excellent abundance estimation accuracy. Although localized exceptions exist where specific baselines yield lower spectral-shape errors—such as A2SAN for tree SAD and the frequency-domain SWC-Net for water SAD—TABG-Net maintains a clear edge in global spatial regularization. This advantage is driven by its transition-aware modeling, which prioritizes resolving subpixel mixing at material interfaces over standalone spectral-shape fitting. Consequently, as illustrated in Figs. 9 and 10, it reconstructs well-aligned endmember curves and abundance maps with sharper boundaries and minimized spatial leakage.

E. Experiments on the Houston Dataset

Table VIII presents the quantitative evaluation on the spectrally complex Houston dataset. Overall, TABG-Net achieves the lowest Mean SAD (0.072) and Mean RMSE (0.068),

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TABLE VII

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE SAMSON DATASET. THE BEST RESULTS ARE IN BOLD

Metric	Class	VCA+FCLS	DAEN	CyCU-Net	DeepTrans	A2SAN	SWC-Net	SRViT	TABG-Net
SAD	Soil	0.027 ± 0.015	0.176 ± 0.004	0.028 ± 0.005	0.036 ± 0.020	0.013 ± 0.003	0.031 ± 0.005	0.024 ± 0.017	**0.009 ± 0.001**
	Tree	0.051 ± 0.004	0.038 ± 0.003	0.041 ± 0.003	0.059 ± 0.015	**0.032 ± 0.001**	0.041 ± 0.000	0.072 ± 0.013	0.035 ± 0.002
	Water	0.133 ± 0.007	0.116 ± 0.014	0.265 ± 0.009	0.063 ± 0.023	0.106 ± 0.007	**0.047 ± 0.012**	0.074 ± 0.022	0.057 ± 0.003
Mean SAD		0.071 ± 0.009	0.110 ± 0.004	0.111 ± 0.004	0.053 ± 0.019	0.050 ± 0.003	0.040 ± 0.004	0.057 ± 0.010	**0.034 ± 0.002**
RMSE	Soil	0.140 ± 0.023	0.195 ± 0.006	0.184 ± 0.019	0.096 ± 0.007	0.070 ± 0.003	0.060 ± 0.004	0.084 ± 0.009	**0.051 ± 0.001**
	Tree	0.094 ± 0.022	0.155 ± 0.002	0.168 ± 0.024	0.073 ± 0.011	0.043 ± 0.005	0.045 ± 0.003	0.089 ± 0.007	**0.042 ± 0.002**
	Water	0.172 ± 0.031	0.074 ± 0.003	0.156 ± 0.022	0.060 ± 0.019	0.053 ± 0.003	0.039 ± 0.004	0.101 ± 0.013	**0.026 ± 0.002**
Mean RMSE		0.135 ± 0.025	0.141 ± 0.002	0.169 ± 0.020	0.076 ± 0.008	0.056 ± 0.003	0.048 ± 0.004	0.091 ± 0.005	**0.040 ± 0.002**

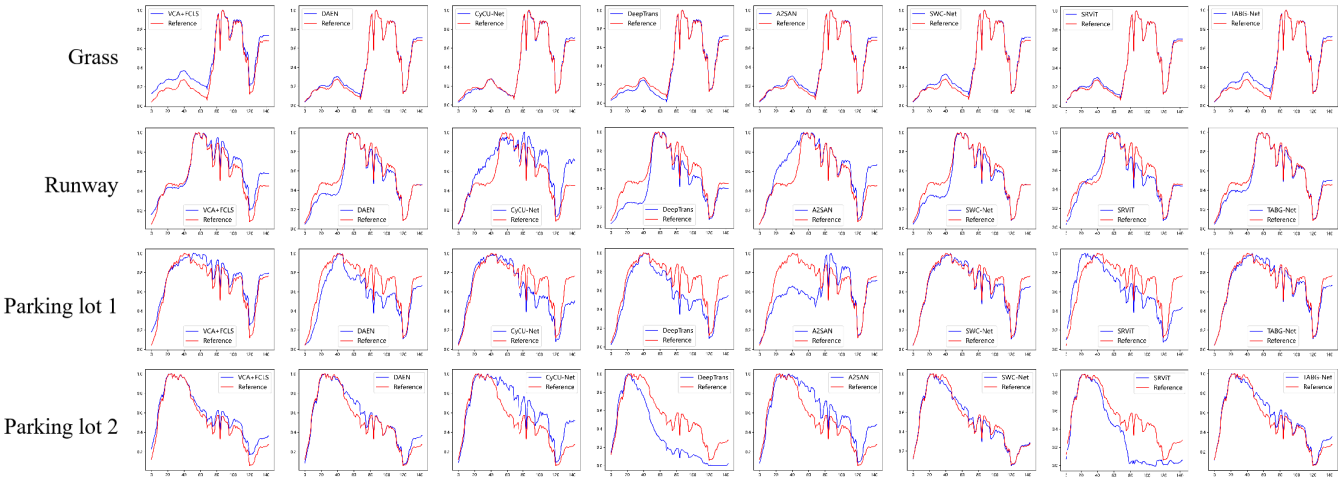


Fig. 11. Visual comparison of the endmembers extracted by different unmixing methods on the Houston dataset.

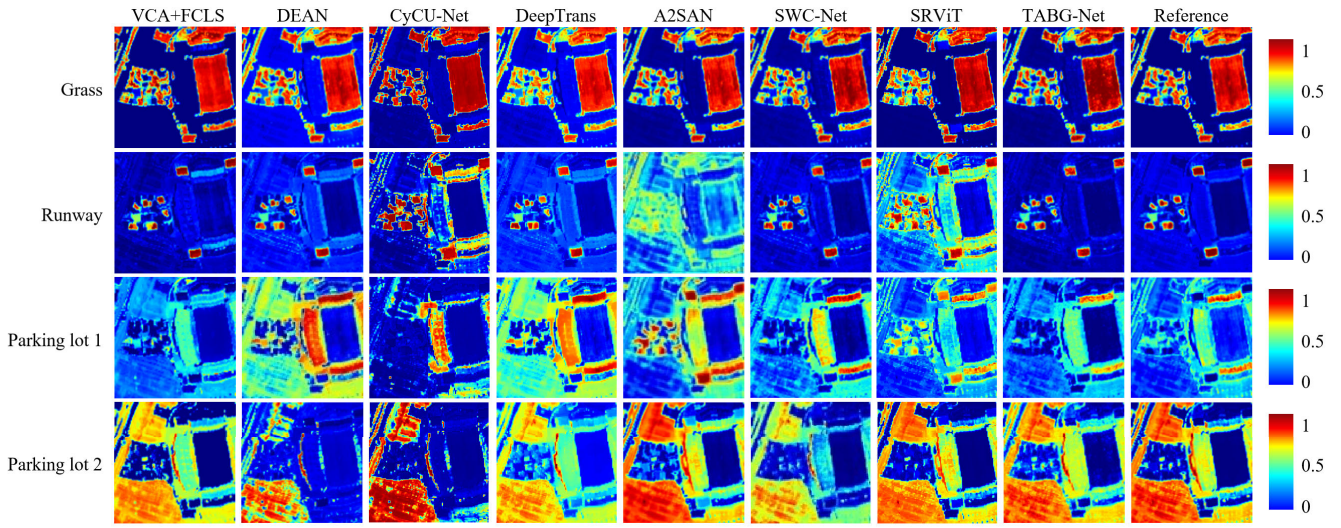


Fig. 12. Visual comparison of the abundance maps estimated by different unmixing methods on the Houston dataset.

demonstrating its superior capability in mitigating material confusion among highly similar urban classes. Compared with advanced spatial–spectral or frequency-domain architectures like SRViT and SWC-Net, TABG-Net exhibits prominent advantages in boundary-dominated categories (such as runways and parking lots) where traditional baselines suffer from boundary leakage. Although localized exceptions exist where alternative methods show marginal advantages—such as CyCU-Net for grass SAD and SWC-Net for runway RMSE—TABG-Net delivers the most robust global unmixing performance. This is confirmed qualitatively in Figs. 11 and 12, where TABG-Net generates well-aligned endmember

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TABLE VIII

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE HOUSTON DATASET. THE BEST RESULTS ARE IN BOLD

Metric	Class	VCA+FCLS	DAEN	CyCU-Net	DeepTrans	A2SAN	SWC-Net	SRViT	TABG-Net
SAD	Grass	0.120 ± 0.026	0.041 ± 0.010	**0.032 ± 0.003**	0.061 ± 0.015	0.041 ± 0.001	0.055 ± 0.001	0.044 ± 0.004	0.081 ± 0.002
	Runway	0.151 ± 0.034	0.130 ± 0.041	0.206 ± 0.032	0.167 ± 0.020	0.206 ± 0.068	0.107 ± 0.004	0.136 ± 0.029	**0.070 ± 0.003**
	Parking lot 1	0.064 ± 0.001	0.170 ± 0.025	0.130 ± 0.027	0.147 ± 0.007	0.180 ± 0.016	**0.061 ± 0.002**	0.102 ± 0.017	0.070 ± 0.007
	Parking lot 2	0.087 ± 0.012	0.096 ± 0.030	0.161 ± 0.026	0.370 ± 0.013	0.384 ± 0.119	0.077 ± 0.004	0.473 ± 0.045	**0.069 ± 0.005**
Mean SAD		0.105 ± 0.012	0.109 ± 0.025	0.132 ± 0.005	0.186 ± 0.013	0.203 ± 0.023	0.075 ± 0.002	0.189 ± 0.014	**0.072 ± 0.007**
RMSE	Grass	0.090 ± 0.005	0.100 ± 0.002	0.076 ± 0.003	0.094 ± 0.004	**0.039 ± 0.001**	0.044 ± 0.004	0.044 ± 0.004	0.055 ± 0.003
	Runway	0.093 ± 0.013	0.078 ± 0.018	0.207 ± 0.004	0.086 ± 0.004	0.244 ± 0.032	**0.038 ± 0.002**	0.126 ± 0.005	0.050 ± 0.003
	Parking lot 1	0.099 ± 0.008	0.090 ± 0.005	0.262 ± 0.055	**0.083 ± 0.003**	0.275 ± 0.050	0.123 ± 0.010	0.123 ± 0.010	0.096 ± 0.004
	Parking lot 2	0.149 ± 0.007	0.169 ± 0.021	0.287 ± 0.034	0.169 ± 0.015	0.223 ± 0.056	0.128 ± 0.012	0.105 ± 0.036	**0.070 ± 0.005**
Mean RMSE		0.109 ± 0.005	0.109 ± 0.011	0.208 ± 0.021	0.108 ± 0.005	0.210 ± 0.028	0.083 ± 0.006	0.099 ± 0.010	**0.068 ± 0.004**

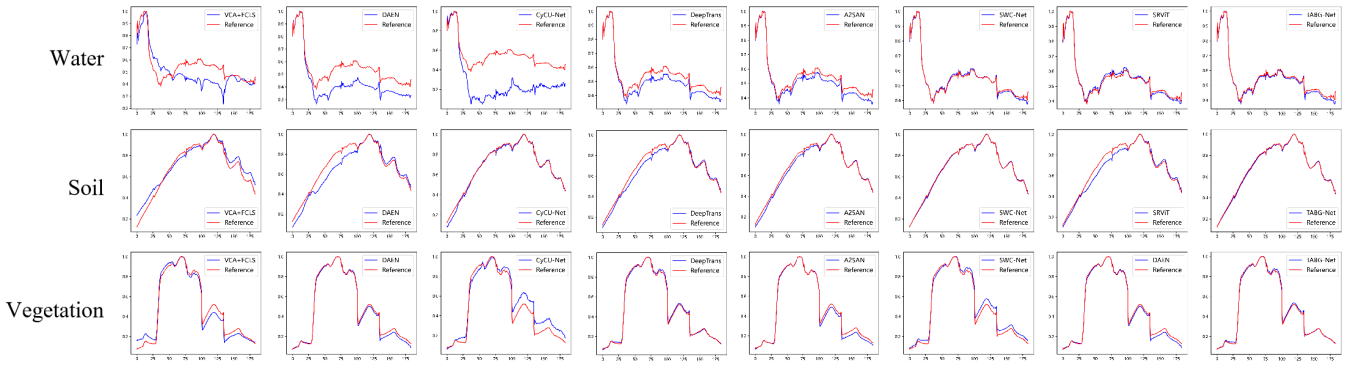


Fig. 13. Visual comparison of the endmembers extracted by different unmixing methods on the Moffett dataset.

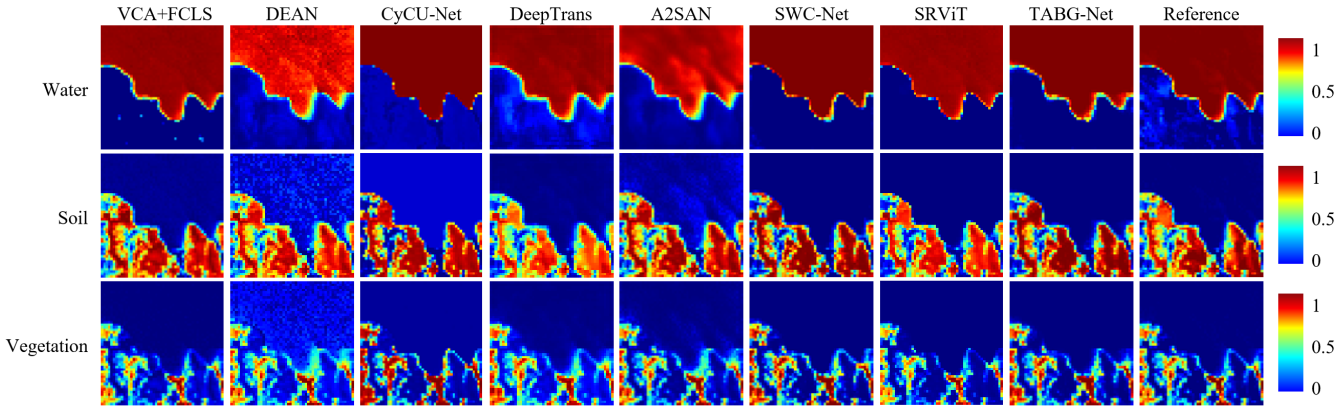


Fig. 14. Visual comparison of the abundance maps estimated by different unmixing methods on the Moffett dataset.

curves and the sharpest abundance maps with minimized spatial leakage.

F. Experiments on the Moffett Dataset

Table IX reports the quantitative results on the complex Moffett dataset. Overall, TABG-Net yields the lowest aSAD (0.016) and Mean RMSE (0.055), demonstrating superior performance under severe endmember variability. Compared to the advanced frequency-domain baseline SWC-Net, TABG-Net maintains a distinct global advantage, particularly across all abundance metrics. Although SWC-Net attains a lower spectral divergence for soil (SAD of 0.008), TABG-Net excels in preserving spatial interface patterns and suppressing background noise across transition zones. This capability is visually corroborated in Figs. 13 and 14, where our model produces tightly aligned spectral curves and highly accurate abundance boundaries.

G. Localized Performance Analysis in Transition Regions

To substantiate the effectiveness of TABG-Net in mitigating error diffusion within material transition zones, we conduct a dual-track localized analysis involving high-resolution abundance profiles and localized quantitative metrics. Unlike global average metrics that may dilute localized improvements, this

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TABLE IX

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE MOFFETT DATASET. THE BEST RESULTS ARE IN BOLD

Metric	Class	VCA+FCLS	DAEN	CyCU-Net	DeepTrans	A2SAN	SWC-Net	SRViT	TABG-Net
SAD	Water	0.187±0.031	0.127±0.015	0.547±0.009	0.038±0.007	0.033±0.004	0.020±0.003	0.041±0.014	**0.018±0.001**
	Soil	0.043±0.005	0.062±0.002	0.036±0.008	0.043±0.002	0.027±0.002	**0.008±0.002**	0.067±0.011	0.011±0.002
	Vegetation	0.103±0.028	0.036±0.005	0.092±0.013	0.023±0.008	0.030±0.009	0.043±0.004	0.031±0.006	**0.018±0.003**
aSAD		0.111±0.020	0.075±0.003	0.225±0.004	0.035±0.005	0.030±0.005	0.023±0.002	0.046±0.010	**0.016±0.002**
RMSE	Water	0.149±0.018	0.173±0.012	0.115±0.005	0.067±0.006	0.106±0.004	0.065±0.005	0.074±0.007	**0.062±0.002**
	Soil	0.128±0.004	0.097±0.007	0.101±0.005	0.082±0.011	0.070±0.005	0.064±0.001	0.102±0.018	**0.059**±0.003
	Vegetation	0.087±0.022	0.115±0.008	0.105±0.004	0.049±0.005	0.059±0.006	0.056±0.004	0.064±0.009	**0.044±0.003**
Mean RMSE		0.121 ± 0.013	0.128 ± 0.009	0.107 ± 0.005	0.066 ± 0.008	0.079 ± 0.004	0.062 ± 0.005	0.080 ± 0.008	**0.055 ± 0.003**

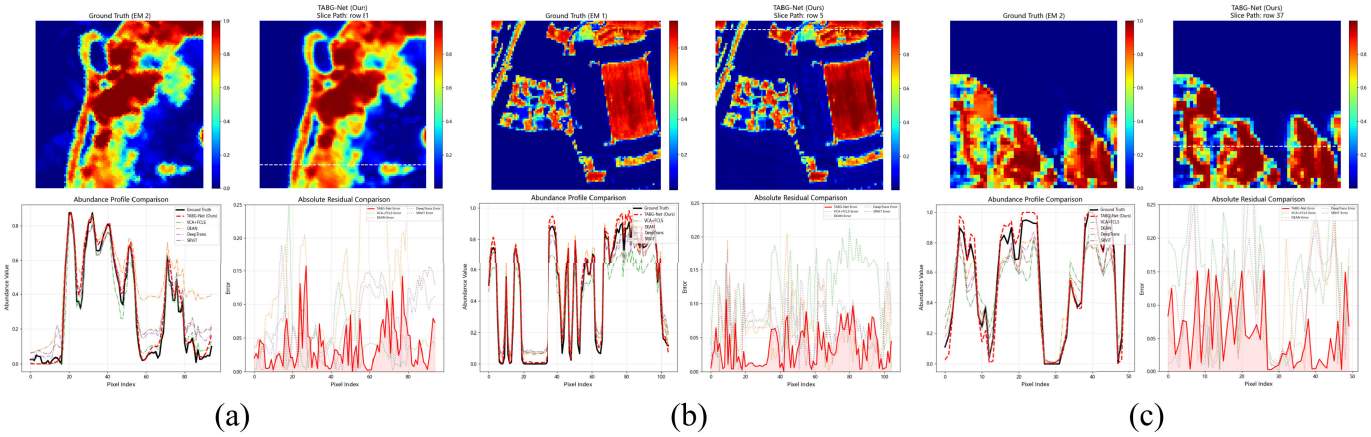


Fig. 15. Localized unmixing performance analysis on real datasets. (a) Samson, (b) Houston, and (c) Moffett. (Top row) GT and TABG-Net abundance maps, with white dashed lines indicating the slice paths across sharp material transitions. (Bottom row) Abundance profile fitting (left) and absolute residual distributions (right) for TABG-Net and four representative methods (VCA-FCLS, DAEN, DeepTrans, and SRVIT). See Section IV-G for detailed analysis.

TABLE X

LOCALIZED RMSE COMPARISON IN TRANSITION ZONES (3-PIXEL BUFFER). THE BEST RESULTS ARE IN BOLD

Method	Samson	Houston	Moffett
VCA+FCLS	0.104	0.098	0.173
DAEN	0.204	0.089	0.148
DeepTrans	0.072	0.103	0.100
SRViT	0.111	0.086	0.123
TABG-Net	**0.045**	**0.078**	**0.085**

approach directly quantifies the model's fidelity at sharp abundance gradients. As illustrated in Fig. 15, we select representative slice paths across the most challenging material boundaries for each dataset. The abundance profiles reveal that while traditional VCA-FCLS and classic AEs like DAEN often produce blurred transitions or significant overestimations, TABG-Net tracks the GT abundance $\mathbf{A}$ with the highest precision. Furthermore, the absolute residual plots quantify the estimation error at each pixel along the path, demonstrating that TABG-Net consistently maintains the lowest residual peaks specifically at the boundary jump points.

TABLE XI

ABLATION STUDY ON NETWORK STRUCTURE (SNR = 30 DB). THE BEST RESULTS ARE IN BOLD

Config	Dual-Freq	Boundary	Ori-CFEM	Mean SAD	Mean RMSE
Baseline	✓	×	×	0.066±0.001	0.069±0.001
+Boundary	✓	✓	×	0.058±0.001	0.062±0.002
+Ori-CFEM	✓	×	✓	0.061±0.002	0.064±0.002
Full-Struct	✓	✓	✓	**0.053±0.002**	**0.056±0.002**

TABLE XII

ABLATION STUDY ON AUXILIARY MODULES (SNR = 30 DB). THE BEST RESULTS ARE IN BOLD

Config	SSEA	SRAM	LERC	Mean SAD	Mean RMSE
Full-Struct + All aux	✓	✓	✓	**0.051±0.002**	**0.056±0.003**
–SSEA	×	✓	✓	0.058±0.002	0.060±0.004
–SRAM	✓	×	✓	0.056±0.002	0.058±0.002
–LERC	✓	✓	×	0.059±0.002	0.060±0.002
–SSEA & –LERC	×	✓	×	0.061±0.004	0.062±0.003

To provide a more granular numerical verification, we calculate the average RMSE within a 3-pixel-wide buffer centered on these transition paths. This localized metric evaluates the model's performance precisely where spectral mixing is most severe, avoiding the subjective bias inherent in broad

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TABLE XIII
EFFECT OF DIFFERENT WAVELET BASIS FUNCTIONS ON TABG-NET. RESULTS ARE REPORTED AS MEAN ± STD OVER THREE RUNS. THE BEST RESULTS ARE IN BOLD

Wavelet	Sim (20 dB)		Samson		Houston		Moffett	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
sym4	**0.063±0.001**	0.069±0.002	0.033±0.003	0.044±0.004	**0.078±0.002**	**0.070±0.001**	0.019±0.003	0.057±0.003
db4	0.065±0.002	0.067±0.003	0.039±0.007	0.043±0.005	0.084±0.005	0.077±0.006	0.017±0.005	0.057±0.005
coif3	0.064±0.005	**0.066±0.001**	0.034±0.003	0.042±0.002	0.089±0.005	0.081±0.003	0.026±0.007	0.061±0.003
bior3.3	0.064±0.002	0.067±0.002	**0.032±0.001**	**0.041±0.001**	0.080±0.007	0.073±0.004	**0.015±0.002**	**0.056±0.002**

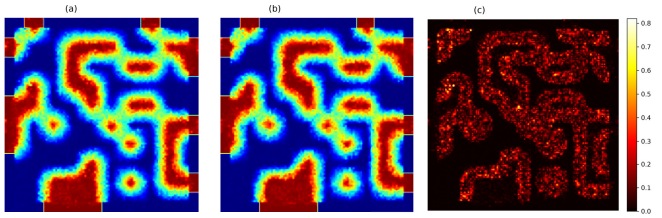


Fig. 16. Visual ablation study of the boundary-guided mechanism on the synthetic dataset under a 30-dB noise level. (a) Abundance map of the full TABG-Net. (b) Abundance map of the ablation variant without BGM. (c) Absolute difference map highlighting the edge-enforcement intensity.

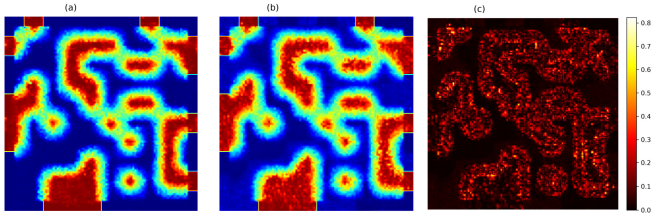


Fig. 17. Visual ablation study of the LERC module on the synthetic dataset under a 30-dB noise level. (a) Abundance map of the full TABG-Net. (b) Abundance map of the ablation variant without LERC. (c) Absolute difference map indicating the localized boundary refinement.

regional partitioning. As summarized in Table X, TABG-Net significantly outperforms all baseline methods in these critical zones. Specifically, on the Samson dataset, TABG-Net achieves a localized RMSE of 0.045, representing an improvement of over 37% compared to the best-performing baseline, DeepTrans (0.072). Similar trends are observed in the Houston and Moffett Field datasets, where the localized error reduction is consistently more pronounced than the global average improvement, directly justifying the transition-aware modeling.

H. Ablation Study

To analyze the impact of different components in TABG-Net, we performed ablation experiments on the synthetic dataset with an SNR of 30 dB, considering both the structural design and auxiliary modules.

1) Structural Ablation: Table XI reports the effect of each major structural element. The baseline adopts only a dual-frequency design, which separates the LF and HF components for independent processing. The addition of the boundary fusion module allows the model to better integrate spatial boundary information, enhancing the delineation of material transitions. Incorporating the Ori-guided CFEM enables targeted modulation of features based on original spectral cues, which improves spectral consistency. Combining both modules in the full structure yielded the most accurate and stable unmixing results, indicating their complementary benefits.

To visually demonstrate this structural impact, Fig. 16 provides a representative qualitative ablation of the boundary-guided modeling fusion mechanism under a 30-dB noise level. As shown in the absolute difference map [see Fig. 16(c)], the abundance error corrections brought by BGM are heavily concentrated along the material interfaces, directly proving that our dual-stream structural design effectively sharpens boundaries and suppresses spatial leakage in transition zones.

2) Auxiliary Module Ablation: Table XII evaluates the contribution of three auxiliary modules—SSEA, SRAM, and LERC—on top of the full structural configuration. The SSEA module reinforces spectral–spatial interactions, improving the representation of both global and local patterns. The SRAM module enhances the network's ability to preserve fine-grained spatial structures, particularly in regions with subtle abundance variations. The LERC module focuses on edge-aware refinement and effectively suppresses artifacts near the boundaries. Removing any single module resulted in a slight performance drop, whereas removing multiple modules led to a more pronounced degradation, highlighting their synergistic effect.

To intuitively verify the localized efficacy of these auxiliary elements, Fig. 17 illustrates a representative visual ablation of the LERC module. It is essential to clarify that while the removal of LERC results in a seemingly subtle increase of 0.004 in global RMSE due to the spatial dilution effect across extensive homogeneous backgrounds, the pixel-wise difference map [see Fig. 17(c)] uncovers a profound localized mechanism. The error corrections are perfectly topology-aligned with the continuous material interfaces, reaching a peak boundary refinement intensity of nearly 0.6 while remaining completely dark in flat interior zones. This empirical evidence firmly refutes the potential risk of noise amplification and confirms that the synergistic interplay of these boundary-targeted components is indispensable to the robust performance of TABG-Net.

3) Sensitivity to Wavelet Bases: In this study, we evaluate the impact of different wavelet basis functions on HU. The

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wavelet functions sym4, db4, coif3, and bior3.3 were compared, with the bior3.3 function being chosen for the proposed method due to its biorthogonal properties. The results, shown in Table XIII, indicate that performance differences across various datasets (Sim (20 dB), Samson, Houston, and Moffett) are minimal. However, the bior3.3 wavelet consistently provides better performance on the Samson dataset, demonstrating its ability to better preserve signal integrity and boundary details. This is particularly beneficial for datasets with complex material transitions and sharp boundaries. While other wavelet bases perform similarly in less complex datasets, bior3.3 proves to be superior in capturing fine structural details in challenging environments, thereby justifying its choice in the proposed method.

V. Conclusion

In this article, we proposed TABG-Net, a transition-aware boundary-guided dual-frequency network for HU that leverages wavelet-based decomposition to better handle challenging transition zones. By integrating complementary LF and HF representations under boundary-aware guidance, TABG-Net improves the accuracy and spatial fidelity of abundance estimation, yielding consistently strong performance on multiple benchmark and real-world datasets in terms of aRMSE and aSAD. In particular, the proposed design enhances the preservation of material boundaries and reduces error diffusion across neighboring regions, which is critical for complex scenes with severe spectral mixing.

Despite these advantages, TABG-Net faces several limitations. First, its performance can be constrained by limited reference endmember diversity in real datasets. Second, the framework operates under a pure linear mixing assumption, leaving room for improvement in highly nonlinear environments. Finally, the current single-level 1-D separable wavelet design can introduce localized directional artifacts or aliasing and lacks multiscale geometric modeling.

Future work will focus on extending this frequency-aware framework toward more realistic unmixing conditions, including endmember variability and nonlinear mixing modeling. Furthermore, we will explore advanced multiscale adaptive frequency representations and stronger spatial constraints to further enhance the model's robustness and generalization.

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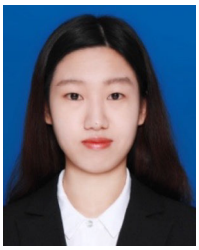
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Yaxiong Chen (Member, IEEE) is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China. His research interests include pattern recognition, machine learning, hyperspectral image analysis, and medical imaging.

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Siqi Liang received the B.S. degree in information engineering from Shanghai Jiao Tong University, Shanghai, China, in 2024, China, where she is currently pursuing the Ph.D. degree in computer science.

Her research interests include action recognition, video understanding, and agent-based multimodal reasoning.



Quanchuan Cheng received the B.S. degree in engineering from Chongqing University of Posts and Telecommunications, Chongqing, China, in 2021, and the M.S. degree from the School of Computer and Artificial Intelligence, Wuhan University of Technology, Wuhan, China, in 2026.

His research interests include hyperspectral image unmixing.



Bo Zhang received the B.S. degree in engineering from Jiangxi Normal University, Nanchang, China, in 2021, and the M.S. degree from the School of Computer and Artificial Intelligence, Wuhan University of Technology, Wuhan, China, in 2024, where he is currently pursuing the Ph.D. degree.

His research interests include pattern recognition, image classification, and multimodal data processing, particularly in remote sensing applications.



Shengwu Xiong received the B.Sc. degree in computational mathematics from Wuhan University, Wuhan, China, in 1987, and the M.Sc. and Ph.D. degrees in computer software and theory from Wuhan University, in 1997 and 2003, respectively.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include intelligent computing, machine learning, and pattern recognition.



Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His research interests include intelligent optical sensing, pattern recognition, machine learning, and hyperspectral image analysis.

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